

Statistics :- →

Statistics is a branch of Mathematics that involves collecting, analysing, interpreting and presenting data.

It provides tools and methods to understand and make sense of large amount of data and to draw conclusions and make decision based on the data.

Use Cases :- →

① Business (DA) → Sales Analysis,
Churn Rate | behaviors
↑
Airtel - idea

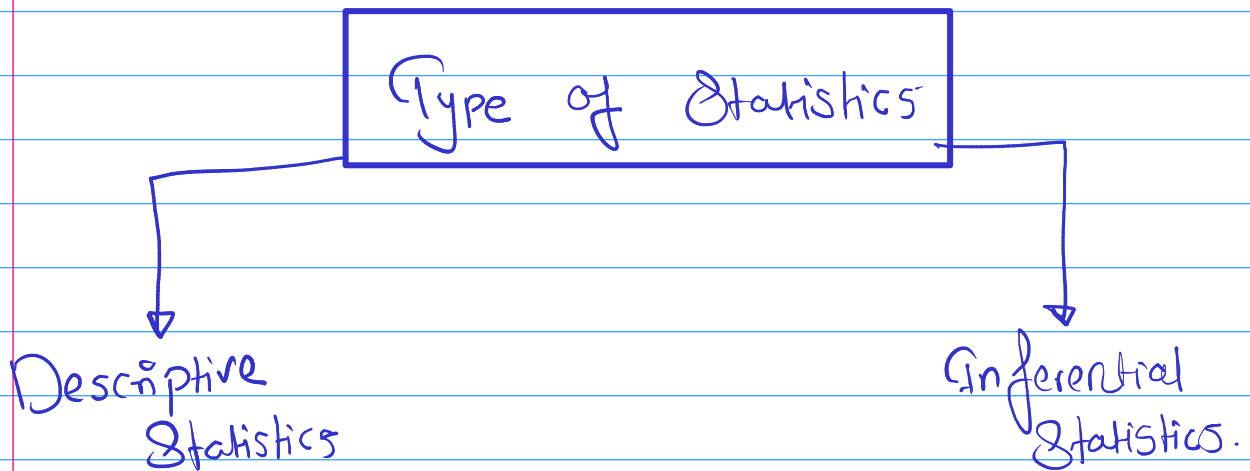
② medical & Drugs formation, clinical trial
(Covid19 - covid shield/
Covaxin)

③ Gov./politics → Polling | Gov. Schemes.

④ Environmental →

Data ? $\Rightarrow$ Facts / pieces of information.

ex - Height of student
 $\rightarrow$ BP
- no. of student.



$\hookrightarrow$ It Deals with Data Analyst.

$\hookrightarrow$ It deals with Data Sci. & AI

$\hookrightarrow$ It deals with the Collection, Organization, analysis, interpretation and presentation of data.

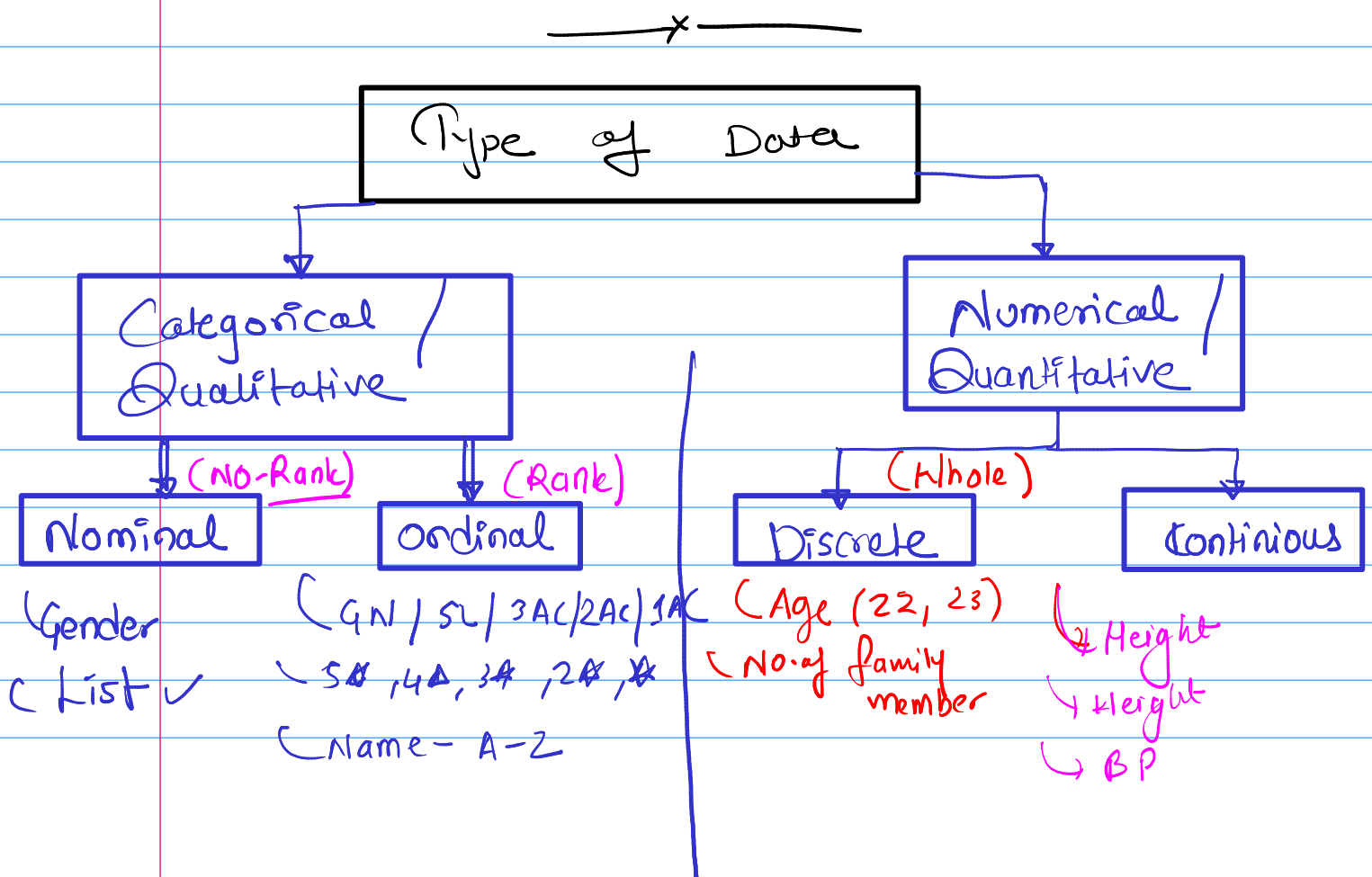
$\hookrightarrow$ It deals with making Conclusion and prediction about a population data based on the Sample data.

$\hookrightarrow$ It involve the use of probability theory to estimate, Hypothesis testing to determine the claims.

Population (N) $\rightarrow$ It refers to the entire group of individuals or object the we are interested in studying.

Sample (n) $\rightarrow$ It is the subset of the population. It is the smaller group of individual or object that we are interested in studying.

ex $\Rightarrow$ All Cricket fans Vs fans who were present in the stadium.



Measure of Central Tendency (Avg)

It is a statistical measure that represents a typical or central value for a dataset.

It provides a summary of the data by identifying a single value that is most representative of the dataset as whole.

① mean

② median

③ mode

① Mean $\Rightarrow$ (Avg) $\Rightarrow$ The mean is the sum of all values in the dataset divided by the number of values.

$$\begin{array}{l} 55 \\ 57 \\ 97 \\ 85 \\ 46 \end{array} \left. \vphantom{\begin{array}{l} 55 \\ 57 \\ 97 \\ 85 \\ 46 \end{array}} \right\} \text{Avg mode?} \quad \frac{55 + 57 + 97 + 85 + 46}{5}$$

$$\text{data} = [1, 2, 3, 4, 5]$$

$$\text{mean} = \frac{1 + 2 + 3 + 4 + 5}{5} = \frac{15}{5} = 3$$

Imp

Note $\Rightarrow$ mean is use to replace the missing values in the dataset if there are no - outliers.

100 student class (12th Pass)

$\infty =$

Age [17, 18, 19, 17, 18, 19, ...] 10 Age missing

$\Rightarrow$ mean is affected by outlier:

10 student data

[17, 18, 19, 18, 17, 19, 18, 18, 19, 17, 17]

$163 + 17$

outlier

mean = 188

— x —

② Median $\Rightarrow$ It the middle no. of the dataset, after sorting the data.

$\Rightarrow$ It is not affected by the outlier

$\Rightarrow$ Always replace missing values with median.

[17, 18, 19, 17, 18, 18, 19, 18, 19, 19, 17, 17]
1 2 3 4 5 6 7 8 9 to

$$\underline{\text{Median}} = \frac{n}{2} = \frac{10}{2} = \boxed{5}$$

$\boxed{18}$

$[7, 8, 17, 18, 19, 18, \dots]$



③ mode → It is the most frequently appearing value in the dataset.

→ It is only use to replace the missing value in the Categorical data.

100 students
9 student Gender is missing

→ 60 → Male → mode
31 → Female

→ Male

M = 45
F = 45

10 m we have two mode.